

Materials Science 2 Worksheet 1 – Nichromes

DR ADIJAT 'WUMI INYANG

A.INYANG@QMUL.AC.UK



Background

Nichrome alloys are the glowing heart of dryers and toasters.

Some everyday products function in an effective way because of the properties of nichromes.



This Photo by Unknown Author is licensed under [CC BY-SA](#)

Aims of project



Introduce students to properties of metal alloys



Conduct a research literature on metal alloys



Establish relationships between properties and applications of metal alloys

Tasks

- ❖ Use literature available in books, research papers and databases to research Nichrome alloys
- ❖ Search information on properties and processing of these materials
- ❖ Find examples and uses of these materials
- ❖ Prepare a report on the structure, composition, properties, processing and uses of Nichrome alloys

Key aspects to research

What are Nichromes? What is the chemical composition and microstructure of these alloys? Could you comment on the type of chemical bond that is present in these alloys?

When where Nichromes described? Could you provide a short summary of the historical development of Nichromes?

What are the main properties of Nichromes? What properties make Nichromes a valuable material? What does service temperature mean and how is this important when discussing the applications of Nichromes?



[This Photo](#) by Unknown Author is licensed under [CC BY-SA](#)

Key aspects to research

Are Nichromes in the 'critical materials' list? What is the definition of a critical material?

Which compounds increase oxidation resistance to Nickel alloys? Perform a short search using databases and information on the internet.

Give at least 2 examples of the use of Nichromes in everyday products that do not include the examples in the description of this project. Could you comment on the economics and environmental impact of using Nichromes?



This Photo by Unknown Author is licensed under [CC BY-NC-ND](#)

Report

2-page report
with findings
that contains
the following
sections

Introduction

Historical development of Nichromes, chemical composition and methods of fabrication

Advantages and disadvantages of Nichromes alloys as materials, including use in everyday products

Summary of report

References

Assessment and Deadline

This is an individual report

It represents 10% of the total mark for this module. The report will be assessed on the technical and English merit. Check your spelling for any mistakes.

Feedback will be provided in QMPlus

The deadline is: 10/03/2023

Please note: We will contact you if there are any problems with uploading coursework to QMPlus and provide alternative means of submission.

DO NOT submit your coursework by email, it will not be considered for marking.